

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026umhackathon@um.edu.my**Project Name:** Homie — AI-Powered Rental Search Aggregator**Version:** 1.0**Date:** April 2026

1. Project Overview

Problem Statement: Renters in Malaysia are forced to manually search across multiple disconnected platforms each with inconsistent listing formats, varying price conventions, and no unified way to compare options. This fragmented landscape wastes hours of a tenant's time, makes fair price assessment nearly impossible, and often results in missed opportunities due to information overload.

Target Domain: Residential rental discovery in Malaysia, with an initial focus on Kuala Lumpur and Klang Valley which are seen as the most active and densely listed urban rental markets in the country.

Proposed Solution Summary: Homie is an AI-powered rental platform that automates multi-platform web scraping across social media. It normalises all collected listings into a consistent data schema, scores each listing against the user's stated preferences using a GLM-5.1 AI scoring engine, and presents ranked results with plain-language match explanations. The platform eliminates repetitive manual searching and empowers tenants to make faster, more confident rental decisions.

2. Background & Business Objective

2.1 Background of the Problem

Malaysia's rental market is highly fragmented. Tenants must visit three or more separate platforms to achieve reasonable market coverage, yet each platform presents listings in structurally incompatible formats. For example, iBilik focuses exclusively on per-room rentals, while iProperty and Facebook cover both whole-unit and room listings without clear distinction. Hence, tenants have no objective way to compare listings according to their priorities. Every decision currently relies on manual, time-consuming cross-referencing.

2.2 Importance of Solving This Issue

Compressing the rental discovery cycle from hours to minutes delivers direct, measurable value to tenants. A scored, ranked, and explained results set means a renter can arrive at a shortlist of 3–5 high-confidence

options in a few minutes rather than 2–3 hours of manual work. The match score rationale also builds trust by making the AI's reasoning transparent and auditable by the user.

At scale, this also improves lead quality for landlords as inquiries generated through Homie come from tenants who have already seen a scored explanation of why the property fits them, resulting in more purposeful and conversion-ready outreach.

2.3 Strategic Fit / Impact

Homie advances **SDG 9 (Industry, Innovation and Infrastructure)** by applying AI infrastructure to modernise a fragmented, inefficient market process that millions of renters navigate daily. Access to adequate housing is a foundational enabler of economic participation, yet rental discovery in rapidly urbanising cities remains stuck in a pre-digital workflow. Homie addresses this without requiring landlords to change behaviour — it builds an intelligent layer on top of existing market data to dramatically improve transparency and efficiency for tenants, particularly fresh graduates, migrant workers, and lower-income households who are most disadvantaged by information fragmentation. The platform's core architecture — an AI orchestration layer that normalises, scores, and explains heterogeneous real-world data — is a replicable model extensible well beyond the rental vertical.

3. Product Purpose

3.1 Main Goal of the System

To provide renters with a single intelligent interface that aggregates rental listings from multiple platforms, normalises them into a consistent schema, scores them against the user's defined preferences, and presents ranked results with transparent explanations, thus, eliminating the need to visit multiple platforms manually.

3.2 Intended Users (Target Audience)

- **Primary Users:** Urban Malaysian tenants actively searching for rental accommodation (rooms or full units) in Kuala Lumpur and the Klang Valley, particularly those with specific requirements around transport, budget, furnishing, and property policies
- **Secondary Users:** Fresh graduates relocating for employment, expatriates unfamiliar with local rental platforms, and working professionals with limited time available for property research
- **Tertiary Users:** Property agents or landlords who wish to benchmark their listings' market competitiveness against aggregated, scored results

4. System Functionalities

4.1 Description

Homie operates as a GLM-5.1 orchestrated agentic pipeline. A renter submits preferences either through a chat-based intake agent or a structured filter form. The backend creates a session and starts a search pipeline driven by a dual-level ReAct orchestrator: a top-level orchestrator decides which pipeline stages to run next, while per-stage sub-agents handle complex reasoning such as field extraction, multilingual translation, and message drafting. Deterministic application code handles scraping, persistence, and numeric scoring. Progress updates are streamed to the frontend in real time through server-sent events, so the user can observe the workflow as it runs.

The pipeline follows this flow:

1. Accept and validate filter input (chat agent or form)
2. Run scraper tools for all four supported sources concurrently
3. Normalize multilingual scraper output into a common listing schema (with GLM translation)

4. Score listings with deterministic backend rules across eight dimensions
5. Generate a short summary report and per-listing explanations
6. Optionally initiate automated Telegram outreach to landlords

4.2 Key Functionalities

- **Chat-Based Search Intake:** An AI chat agent accepts natural language input and extracts structured filters with per-field confidence tracking (confirmed / inferred / soft / missing). The agent suggests action chips for quick refinement and signals when enough information is available to start a search.
- **Structured Filter Form:** Users can alternatively submit a nine-field form covering location, price range, room type, furnishing preference, gender restriction, transport preference, parking requirement, pet-friendly preference, and max results.
- **Multi-Source Listing Collection:** The backend scrapes listings from ibilik, iProperty, PropertyGuru, and Facebook. If one source fails, the system continues with partial results from remaining sources.
- **Deterministic Match Scoring:** Each normalized listing receives a 0–100 score across eight dimensions: price fit, location match, room type, transport proximity, furnished status, parking availability, pet policy, and gender restriction. Unknown fields receive partial scores rather than zero so sparse listings are not unfairly penalized.
- **Explainable Ranked Results:** Listings are shown in descending score order by default. The system generates short plain-language explanations per listing and a session summary report. Users can sort by best match, price low-to-high, or price high-to-low, and filter by source when multiple sources return results.
- **Automated Telegram Outreach:** Users can initiate automated multi-turn Telegram conversations with landlords. The system resolves contact details, generates a professional opening inquiry via GLM, sends it via the Telegram MTPROTO API, and tracks the ongoing conversation state including verification of must-have requirements.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

Model used: Z AI GLM-5.1 provided by YTL AI Labs.

GLM-5.1 was selected as the core LLM for the following reasons:

1. It provides native support for structured tool/function calling, which is essential for the Orchestrator's multi-step agent invocation pattern.
2. It demonstrates strong instruction-following fidelity for both structured JSON output (normalisation and scoring steps) and natural language generation (inquiry drafting and report generation).

4.3.2 Prompting Strategy

Homie uses a dual-level ReAct (Reason + Act) agentic prompting strategy.

A top-level orchestrator maintains a ReAct loop across the full pipeline, choosing the next tool to invoke from nine available workflow tools based on current session state and prior tool results.

Per-stage sub-agents run bounded ReAct loops (max ten iterations each) within complex pipeline stages such as filter validation, normalization, report generation, and outreach drafting.

A separate chat intake agent extracts structured filters from natural language with confidence annotations and suggested chips.

Deterministic rules are used for numeric scoring so the model explains the ranking rather than inventing it.

This hybrid strategy was chosen because the system benefits from model-driven control flow and language generation, but the highest-risk tasks (scoring, persistence, scraping) are safer when implemented as code.

4.3.3 Context & Input Handling

Homie keeps prompt sizes manageable with explicit limits:

Search filters are passed as a compact JSON-like payload rather than prose.

Per-source scraping is capped by MAX_LISTINGS_PER_SOURCE (default: 25).

Long listing descriptions are trimmed before prompt use. Outreach drafting uses shortened descriptions; score explanation prompts use compact summaries.

Normalization processes listings in bounded batches to keep per-call token counts stable.

If inputs exceed practical limits, the system truncates listing payloads and continues with the bounded subset so the workflow can complete.

4.3.4 Fallback & Failure Behaviour

Homie is designed to degrade gracefully rather than fail hard:

If a model call fails, the client retries once after a short delay.

If a scraper fails, the orchestrator continues with remaining sources and returns partial results.

If score explanation generation fails, the system keeps a rule-based fallback explanation.

If summary report generation fails, the system falls back to a short template report.

If inquiry generation fails, the system returns a safe boilerplate inquiry draft.

If the orchestrator hits its maximum iteration limit, it emits an error event but does not crash the session.

The product remains usable even under unstable model responses, source throttling, or partial scraper outages.

5. User Stories & Use Cases ---

5.1 User Stories

1. Ali is a worker moving to kl for work, but he doesn't have time looking through different platforms and doesn't know a lot about the strategic locations in kl.
2. Sarah, a 2nd year university student and 4 of her friends are looking for a room, all of them have different budgets, and send different links from platforms and couldn't come up with a summary.
3. Lee found many rental posts on facebook and XiaoHongShu, but they require him to manually message the agents to check for availability.

5.2 Use Cases (Main Interactions)

UC-01 - Chat Intake: User types their requirements in the chat interface. The chat agent extracts filters, shows confidence indicators, suggests refinement chips, and prompts the user to start the search when ready.

UC-02 - Form Intake: User fills in the nine-field filter form and submits directly. The backend creates a new session and starts the pipeline.

UC-03 - Watch Pipeline Progress: While the backend runs, the workflow page receives server-sent events and shows step-by-step status updates for validation, scraping (four sources), normalization, scoring, and report generation.

UC-04 - Review Ranked Results: Once the pipeline completes, the user reviews the summary report and listing cards, sorts by best match or price, and filters by source if needed.

UC-05 - Initiate Telegram Outreach: User clicks "Start Telegram Outreach" on a listing. The system generates an opening message, sends it to the landlord's Telegram, and tracks the ongoing conversation.

UC-06 - Manual Outreach Fallback: User opens the outreach modal for a listing without Telegram contact data. The system drafts a message the user can copy and send via phone or WhatsApp.

UC-07 - View Original Listing: User clicks through to the source listing URL to inspect the original before making contact.

UC-08 - Connect Facebook Session: User clicks "Connect Facebook" on the results page, completes a guided login in the browser, and the system saves the session for future scraping.

UC-09 - Adjust Search and Rerun: User returns to chat or form with previous filters pre-filled, changes one or more values, and starts another search session.

6. Features Included (Scope Definition)

- Web-based rental search dashboard with a chat intake page and a manual filter form page
- Chat-based search intake with conversational filter extraction, per-field confidence tracking (confirmed / inferred / soft / missing), suggested refinement chips, and ready-to-search signaling
- Manual search filter form with nine fields: location, min price, max price, room type, furnished status, gender restriction, transport preference, parking requirement, pet-friendly preference, and max results
- Backend search sessions with a unique session ID and persisted search results with a 24-hour TTL
- Multi-source scraping support for ibilik (REST API), iProperty (Playwright), PropertyGuru (Playwright), and Facebook (Playwright with saved session)
- Facebook session connection flow via guided browser login to enable Facebook group and Marketplace scraping
- GLM-powered multilingual normalization — field extraction and Bahasa Malaysia / Chinese to English translation for listings from all four sources
- URL-based duplicate removal during normalization
- Session progress feed delivered through server-sent events with replay buffer for late-joining clients
- Deterministic 0–100 listing scoring with eight dimensions: price, location, room type, transport, furnished status, parking, pet policy, and gender restriction
- Partial-credit scoring for unknown fields (no harsh zero penalties for sparse listings)
- Generated score explanations per listing and a generated session summary report
- Sort controls for best match, price low-to-high, and price high-to-low on the results page
- Source filtering on the results page when multiple sources return results
- Automated Telegram outreach via Telethon MTPROTO — multi-turn conversation management, must-have verification questions, conversation history tracking
- Manual outreach fallback with GLM-drafted inquiry message when Telegram is unavailable
- Telegram deep link handoff when a Telegram handle is available
- Demo seed mode using fixture JSON for all four sources when live scraping is unreliable
- Incremental listing persistence during pipeline execution (results available mid-pipeline)

7. Features Not Included (Scope Control)

- User accounts, login, saved searches, or persistent renter profiles across sessions
- Fully autonomous landlord reply management or negotiation on behalf of the user
- Ongoing listing monitoring, alerts, or scheduled search automation
- Property booking, payment, or transaction handling
- Native mobile applications

- Map-based search, route planning, or precise geospatial commute scoring
- Full semantic duplicate resolution beyond current URL-based deduplication
- Support for additional platforms such as Xiaohongshu or Mudah in the current prototype
- Full multilingual product UI for the renter (the interface is English-only; multilingual support applies to listing data normalization only)
- Advanced scoring on lower-coverage attributes such as parking and pet policy as primary ranking dimensions in the current prototype

8. Assumptions & Constraints

8.1 LLM Cost Constraint

Estimated model usage for a typical session is in the low tens of thousands of tokens, depending on how many listings are processed and whether the user requests outreach drafting. The main cost-control decisions are:

- capping results per source before they enter downstream prompts (MAX_LISTINGS_PER_SOURCE, default 25),
- keeping numeric scoring deterministic in code rather than asking the model to evaluate every listing,
- truncating long listing text before sending it to explanation or outreach prompts,
- generating one compact summary report instead of verbose narrative output, and
- running per-stage sub-agents with bounded iteration limits (max ten per stage).

8.2 Technical Constraints

- Scraper Fragility: The live pipeline depends on third-party site structures that may change without warning. PropertyGuru and iProperty both use Cloudflare protection that may require browser-based rendering.
- Anti-bot behavior: iProperty, PropertyGuru, and Facebook flows are sensitive to browser automation, throttling, and login walls.
- Facebook authentication dependency: The Facebook scraper requires a saved browser session obtained through the guided login flow. Listings from Facebook are unavailable if the session has expired or was never established.
- Telegram activation dependency: Automated outreach requires TELEGRAM_API_ID, TELEGRAM_API_HASH, and TELEGRAM_PHONE to be configured. Without these, outreach falls back to manual draft-only mode.
- Field coverage variance: Many listing sources do not provide complete structured fields, so some attributes remain unknown after normalization.
- Prototype architecture: Session state is managed primarily for hackathon-scale use, not for large multi-tenant production load.
- Environment configuration: Model endpoint, model name, database connection, Telegram credentials, and demo behavior all depend on environment settings that must be configured before a live run.

8.3 Performance Constraints

Target end-to-end completion time is roughly 45 to 120 seconds depending on source responsiveness, Playwright load time for JavaScript-heavy sites, normalization translation volume, and model latency. The user experience is designed around visible progress updates so that the wait remains understandable. The

system may return partial results if one source is slow or blocked. In a demo setting, the seeded fixture mode is the safety net when live scraping conditions are unstable.

8.4 User Input Assumptions

A valid search requires at least a location, a minimum price, and a maximum price. Other filters are optional and default to broad values. Chat intake is the primary recommended entry point. The structured form provides a fallback for users who prefer explicit fields. Generated inquiry text is a drafting aid. When using manual fallback, the user is expected to review and send the message themselves. Automated Telegram outreach sends on the user's behalf once confirmed. The renter remains responsible for verifying listing accuracy on the original source before committing to contact or payment.

9. Risks & Questions Throughout Development

Risk / Question	Details & Mitigation	Severity
Scraper Reliability	DOM/UI changes on ibilik, iProperty, or Facebook will silently break data ingestion with no immediate user-visible error. Mitigation: implement schema validation at the scraper output boundary; if fewer than 3 required fields are populated for a listing, discard the listing and flag the source as degraded in the pipeline log.	High
Score Inconsistency	GLM-5.1 may produce marginally different scores for semantically identical listings across separate pipeline runs, undermining user trust in rankings. Mitigation: the scoring prompt uses a fixed rubric with defined dimension weights (location 35%, price 30%, transport 15%, contact info 10%, policy match 10%) to constrain output variance.	Medium
Hallucinated Listing Fields	The normalisation agent may fabricate field values not present in the raw scraped data (e.g. inventing a transport station name). Mitigation: the normalisation schema enforces null-allowed fields; the model is explicitly instructed to return null for absent data rather than infer a value. The scoring agent is then instructed to penalise null fields.	High
Facebook Marketplace Degradation	Facebook's anti-scraping countermeasures may result in zero results from this source in certain environments or after IP-level flagging. The pipeline is designed to degrade gracefully to two-source results. Users are notified in the pipeline log.	Medium
Legal & ToS Exposure	Automated scraping may conflict with the Terms of Service of one or more source platforms. Mitigation: Homie operates at human-browsing request rates, does not persist scraped data beyond the session, and does not reproduce full listing content. This is a hackathon prototype and not a production deployment.	Medium
Inquiry Message Quality	GLM-5.1 may generate generic or contextually inappropriate inquiry drafts if listing data is sparse or malformed. Mitigation: the inquiry prompt enforces a minimum context requirement; if fewer than 3 normalised fields are available for a listing, the inquiry preparation feature is disabled for that card with a visible indicator.	Low
Pipeline Timeout	On slow networks or during platform rate-limiting, the full pipeline may exceed the 45-second target, degrading user experience. Mitigation: a visible timeout warning is displayed at 40 seconds with an option to retrieve partial results already available from completed pipeline steps.	Low